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Extract from a Letter to Mr C. W. Merrifield.

[From the Messenger of Mathematics, vol. I. (1872), pp. 87, 88.]

THE general integral of the equations

$$\frac{p}{x} = \frac{dt}{a} = \frac{dq}{y}.$$

$\left[\text{where } a,\, \beta,\, \gamma,\, \delta = \frac{dq}{dx}, \frac{dq}{dy}, \frac{dq}{ds}, \frac{dq}{dt},\right]$ can, I think, be found, viz. $\frac{p}{x} = \frac{dt}{a} = \frac{dq}{y}$
$s =$ function x, and $\frac{dt}{ds} = \frac{a}{y}$ gives $s =$ function t. But $s =$ function x, is integrated as the equation of a developable surface (p instead of s), we must have

$$p = ax + by + p',$$

$$0 = ax + y + p';$$

similarly, $s =$ function t, gives

$$\left(a'\frac{dx}{ds}, \quad y'\frac{dq}{dt}\right);$$

a and p functions of t, and

$$\left(a' = \frac{dx}{ds}, \quad y' = \frac{dq}{dt}\right);$$

Observe that the constants here have been so taken, that $\frac{dq}{ds} = a$, $\frac{dq}{dt} = b$; in order that δ may, in the two pairs of equations, mean the same function of (x, y), we must have

$$s' = \frac{b}{a} - \frac{p}{x}.$$

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that is,

$$b = \int \frac{dx}{y}, \quad t' = \int \frac{dq}{x};$$

or, writing $a = dt$, $p = q$, we have

$$p = abh + pb + q,$$

$$q = bs + p \int \frac{dx}{y} - p \int \frac{a}{y}\, dx,$$

where

$$abh + y + qh = 0.$$

The last equation gives b as a function of (x, y), and the values of p, q are then such that $dx + pdx + qdy$ is a complete differential, so that we obtain z by the integration of that equation.

A simple example is

$$p = \tfrac{1}{2}bs - bq, \quad q = -bs + g \log b, \quad bs - y = 0,$$

that is,

$$p = -\tfrac{1}{4}\frac{y^2}{x}, \quad q = -y + y \log \frac{y}{x},$$

whence

$$z = \tfrac{1}{4}y^2 \log \frac{y}{x} - \tfrac{1}{4}y^2;$$

we have

$$r = \tfrac{1}{2}\frac{y^2}{x^2}, \quad s = -\frac{y}{x}, \quad t = \log \frac{y}{x},$$

$$a = -\frac{y^2}{2x^2}, \quad \beta = -\frac{y}{x}, \quad \gamma = -\frac{1}{y}, \quad \delta = \frac{1}{x};$$

or

$$\frac{a}{\beta} = \frac{p}{x} = \frac{b}{y} \left(= a - \frac{p}{x}\right);$$

as it should be.

Cambridge, 28 July, 1871.

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Further Note on Lagrange's Demonstration of Taylor's Theorem.

[From the Messenger of Mathematics, vol. I. (1872), pp. 105, 106.]

This refers to a paper, Wilkinson, "Note on Taylor's Theorem," *Messenger of Mathematics*, same volume, pp. 96, 97, discussing the "Note on Lagrange's Demonstration of Taylor's Theorem," [248, ante, pp. 493—495].

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On a Property of the Torse Circumscribed about Two Quadric Surfaces.

[From the Messenger of Mathematics, vol. I. (1872), pp. 111, 112.]

THE property mentioned by Mr Townsend in his paper(*) in the August No., "On a Property in the Theory of Confocal Quadrics," may be demonstrated in a form which, it appears to me, better exhibits the foundation and significance of the theorem.

Starting with two given quadric surfaces, the torse circumscribed about these touches each of a singly infinite series of quadric surfaces, any two of which may be used (instead of the two given surfaces) to determine the torse; in the series are included four conics, one of them in each of the planes of the self-conjugate tetrahedron of the two given surfaces; and if we attend to only two of these conics, the two conics are in fact any two conics whatever, and the torse is the circumscribed torse of the two conics; or, what is the same thing, it is the envelope of the common tangent-planes of the two conics.

Consider new two conics U, U', the planes of which intersect in a line I; and let I meet U in the points L, M, and meet U' in the points L', M': take L the pole of I in regard to the conic U, and L' the pole of I in regard to the conic U'.

Take T any point on I, and draw TP touching U in P, and TP' touching U' in P': the points P, P' may be considered as corresponding points on the two conics.

Join AP and produce it to meet the line I in G; the line APG is in fact the polar of T in regard to the conic U (for T being a point on I, the polar of T passes through A; and this polar also passes through P); that is, the points T, G, and L, M are harmonics on the line I; whence also, in the plane of the conic U',

(*) *Messenger of Mathematics, same volume, pp. 48, 49.*

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the lines PG, PT and PL, PM are harmonic lines through the point P. It thus appears that in the particular case where the points L, M are the foci of the conic U', the line PG is the normal at the point P; and we may say in general that PG is the quasi-normal at the point P of the conic U'.

[Figure: Conic with quasi-normal construction — diagram omitted]

Consider now the torse circumscribed about the conics U, U'; the plane PTP' will represent any plane, and the line PP' any line of this torse; projecting on the plane of U' with the point A as centre of projection, the projection of PP is the line PG; which, as just seen, is the quasi-normal of the conic U' at the point P'.

The projection of the cuspidal curve is the envelope of line PG, which is the projection of the generating line PP' of the torse—viz. this envelope is the quasi-evolute of the conic U'; which is the theorem in question.

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On the Reciprocal of a Certain Equation of a Conic.

[From the Messenger of Mathematics, vol. I. (1872), pp. 120, 121.]

THE following formula is useful in various problems relating to conics: the reciprocal equation of the conic

$$\lambda(px+qy+cz)(x+by+dz)+\mu(a''x+b''y+c''z)(a''x+b''y+c''z)=0$$

may be written indifferent in either of the forms

$$|\lambda\,\xi\,,\eta\,,\zeta\;=a\,\xi\,,\eta\,,\zeta\,,\,b\,\xi\,,\eta\,,\zeta\;|{+}4\lambda\mu\,\xi\,,\eta\,,\zeta\,,\,a'\,\xi\,,\eta\,,\zeta\,,\,b'\,\xi\,,\eta\,,\zeta\;|{+}\,|a'\,\xi\,,\eta\,,\zeta\,,\,b'\,\xi\,,\eta\,,\zeta\,,\,c'\,\xi\,,\eta\,,\zeta\;|=0,$$

and

$$|\lambda\,\xi\,,\eta\,,\zeta\;=a\,\xi\,,\eta\,,\zeta\,,\,b\,\xi\,,\eta\,,\zeta\;|{+}4\lambda\mu\,\xi\,,\eta\,,\zeta\,,\,a\,\xi\,,\eta\,,\zeta\,,\,b''\,\xi\,,\eta\,,\zeta\;|{+}\,|a''\,\xi\,,\eta\,,\zeta\,,\,b''\,\xi\,,\eta\,,\zeta\,,\,c''\,\xi\,,\eta\,,\zeta\;|=0.$$

In fact, in the reciprocal equation, noting the coefficient of β, it is

$$\{\lambda(bc'-b'c)+\mu(b''c'-b''c')\}^2=(Dab'-2ab'b'')(Dac'-2bc'c'),$$

viz. this is

$$\lambda^2(bc'-b'c)^2+\mu^2(b''c'-b''c')^2+2\lambda\mu\left\{\begin{array}{l}2bb'c''c''+2b''b'cc'\\-(bc+b'c')(b''c''+b''c'')\end{array}\right\}.$$

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or, as it may be written,

$$[\lambda(bc' - b'c) \pm \mu(b''c' - b''c')]^2 = D\lambda\mu \left\{ \begin{array}{l} 2bb'c''c'' + 2b''b'cc' \\ -(bc + b'c')(b''c'' + b''c'') \end{array} \right\}.$$

Taking the upper signs, this is

$$[\lambda(bc' - b'c) + \mu(b''c' - b''c')]^2 + 4\lambda\mu \left\{ \begin{array}{l} bb'c''c'' + b''b'cc', \\ -bc'b''c'' - b'cb''c'' \end{array} \right\},$$

viz. the term in $\lambda\mu$ is

$$+4\lambda\mu(bc'' - b''c)(b'c' - b'c''),$$

Taking the lower signs, it is

$$[\lambda(bc' - b'c) - \mu(b''c' - b''c')]^2 + 4\lambda\mu \left\{ \begin{array}{l} bb'c''c'' + b''b'cc', \\ -bc'b''c'' - b'cb''c'' \end{array} \right\},$$

viz. the term in $\lambda\mu$ is

$$+4\lambda\mu(bc' - b''c)(b'c'' - b''c').$$

And it is hence easy to infer the forms of the other coefficients, and to arrive at the foregoing result.

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Further Note on Taylor's Theorem.

[From the Messenger of Mathematics, vol. I. (1872), p. 137.]

THIS paper refers to a "Further Note on Taylor's Theorem," *Messenger of Mathematics*, same volume, pp. 135—137.

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On an Identity in Spherical Trigonometry.

[From the Messenger of Mathematics, vol. I. (1872), p. 145.]

In a spherical triangle, writing for shortness α, β, γ for the cosines and α', β', γ' for the sines, of the sides: also

$$\Delta^2 = 1 - \alpha^2 - \beta^2 - \gamma^2 + 2\alpha\beta\gamma;$$

we have

$$\cos A = \frac{\alpha - \beta\gamma}{\beta'\gamma'}, \qquad \sin A = \frac{\Delta}{\beta'\gamma'},$$

with the like expressions in regard to the other two angles B, C respectively.

Hence

$$\begin{aligned}\cos(A+B+C) &= \cos A \cos B \cos C - \cos A \sin B \sin C - \&c. \\ &= \frac{(\alpha - \beta\gamma)(\beta - \gamma\alpha)(\gamma - \alpha\beta) - \Delta^2(\alpha + \beta + \gamma - \beta\gamma - \gamma\alpha - \alpha\beta)}{(1-\alpha^2)(1-\beta^2)(1-\gamma^2)}.\end{aligned}$$

The numerator is identically

$$= (1-\alpha)(1-\beta)(1-\gamma)\{\Delta^2 - (1+\alpha)(1+\beta)(1+\gamma)\},$$

viz. comparing the two expressions, we have

$$\begin{aligned}&(1-\alpha)(1-\beta)(1-\gamma)\Delta^2 - (1-\alpha^2)(1-\beta^2)(1-\gamma^2) \\ &\quad = (\alpha - \beta\gamma)(\beta - \gamma\alpha)(\gamma - \alpha\beta)\Delta^2(-\alpha - \beta - \gamma + \beta\gamma + \gamma\alpha + \alpha\beta);\end{aligned}$$

or, what is the same thing,

$$(1 - \alpha\beta\gamma)\Delta^2 = (1-\alpha^2)(1-\beta^2)(1-\gamma^2) + (\alpha - \beta\gamma)(\beta - \gamma\alpha)(\gamma - \alpha\beta),$$

which is the identity in question and can be immediately verified. We have thus

$$\cos(A+B+C) = \frac{\Delta^2 - (1+\alpha)(1+\beta)(1+\gamma)}{(1+\alpha)(1+\beta)(1+\gamma)},$$

and thence

$$1 + \cos(A+B+C) = \frac{\Delta^2}{(1+\alpha)(1+\beta)(1+\gamma)},$$

$$1 - \cos(A+B+C) = \frac{2(1+\alpha)(1+\beta)(1+\gamma) - \Delta^2}{(1+\alpha)(1+\beta)(1+\gamma)},$$

giving at once the values of $\cos^2 \frac{1}{2}(A+B+C)$, $\sin^2 \frac{1}{2}(A+B+C)$, $\sin(A+B+C)$, and $\tan^2 \frac{1}{2}(A+B+C)$: these are known expressions in regard to the spherical excess.

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On a Penultimate Quartic Curve.

[From the Messenger of Mathematics, vol. I. (1872), pp. 178—180.]

I HAVE had occasion to consider with some particularity the form of a curve about to degenerate into a system of multiple curves; a simple instance is a trinodal quartic curve about to degenerate into the form $x^2y^2 = 0$, or say a "penultimate" of $x^2y^2 = 0$. To fix the ideas, take x, y, z to denote the perpendiculars on the sides of an equilateral triangle, altitude $= 1$ (so that $x + y + z = 1$), and let the curve be symmetrical in regard to the coordinates x, y, its equation being thus

$$(a, a, 1, f, f, h)(x, y, z)^2 = 0,$$

where a, f, h are ultimately all indefinitely small in regard to unity: to diminish the number of cases I further assume

$$a = +, \quad f \text{ and } h = -,$$

$$h^2 > a^2, \quad \text{that is, } a + h = -,$$

$$f^2 > a, \quad \text{,, ,, } \sqrt{(a)} + f = -,$$

but I do not in the first instance take a, f, h to be indefinitely small. Then if $-f$ is not too large, the curve is as shown in the figure[1], viz. it is a triloop curve, with two horizontal double tangents, 3 touching the curve in two real points, 4 touching it in two imaginary points. Imagine $-f$ increased: the new curve will have the same general form, intersecting the first curve at A and B but touching it at C, viz. it will pass inside the loop C but outside the loops A, B; and outside the remainder of the curve; and the 4 will also move downwards as shown. The new position of 4 will be below the first position.

Supposing that a, h have given values, and that $-f$ continually is increased in regard to $\sqrt{(a)}$; two things may happen. First, the double tangent 3 may move down

[1]The figure is drawn with very small values of a, f, h, in order to exhibit as nearly as may be one of the penultimate forms of the curve; but this is not in anywise assumed in the reasoning of the text. Observe in the figure that the points A, B are ordinary double points, and that there are at each of them two distinct tangents inclined at a small angle to each other.

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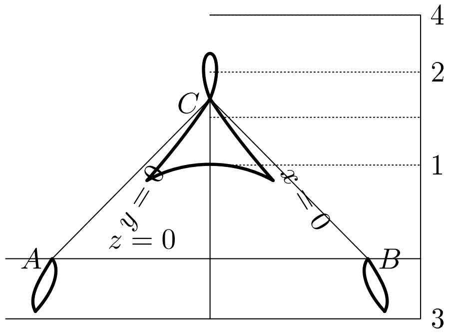
to $z = -\infty$, the lower loops lengthening out and finally becoming each of them a pair of parabolic branches parallel at infinity; and then reappearing at $z = +\infty$, again move downwards, each loop becoming in this case a pair of hyperbolic branches touching two asymptotes at $z = -\infty$, and then again on the opposite sides thereof at $z = +\infty$,



and coming down as a single branch to touch the double tangent 3 which is now above 4. Secondly, the double tangent 4 may come to coincide with the horizontal tangent 2, at the instant of coincidence being a tangent of four-pointic contact; and becoming afterwards (being as before above 2) an ordinary double tangent with two real points of contact; viz. instead of a simple loop at C we have a heart-shaped loop.

But to investigate whether the two cases actually happen, and in what order of succession, we require the expressions of z for the several lines in question; we find, without difficulty,

$$\begin{array}{rrcll}
\text{for line } 1, & z_1 & = & \dfrac{1}{1+2\lambda_1}, & \text{where} \quad \lambda_1 = -2f + \sqrt{4f^2 - 2(a+h)}, \\
2, & z_2 & = & \dfrac{1}{1-2\lambda_2}, & \lambda_2 = 2f + \sqrt{4f^2 - 2(a+h)}, \\
3, & z_3 & = & \dfrac{1}{1-2\lambda_3}, & \lambda_3 = \dfrac{a-h}{2\{-\sqrt{(a)} - f\}}, \\
4, & z_4 & = & \dfrac{1}{1-2\lambda_4}, & \lambda_4 = \dfrac{a-h}{2\{\sqrt{(a)} - f\}},
\end{array}$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are all positive. Observe that in the limiting case $-f = \sqrt{(a)}$, where, instead of the loops at A, B, we have cusps; z_1, z_2, and z_4 are (in general)

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positive; $\lambda_3 = \infty$, and therefore $z_3 = 0$; that is, the line 3 coincides with AB, ceasing to be a double tangent; there is in this case the one double tangent 4.

First. z_3 becomes infinite for $1-2\lambda_3 = 0$; that is, $a-h = -\sqrt{(a)}-f$, or $-f = \sqrt{(a)}+(a-h)$; viz. for $-f = \sqrt{(a)} + (a-h) - \varepsilon$, we have $z_3 = -\infty$, and for $-f = \sqrt{(a)} + (a-h) + \varepsilon$, we have $z_3 = +\infty$.

Secondly. The lines 4 and 2 will coincide if

$$\lambda_4 \left[= \frac{a-h}{2\{\sqrt{(a)} - f\}} \right] = 2f + \sqrt{4f^2 - 2(a+h)},$$

that is, if

$$\lambda_4(\lambda_4 - 4f) = -2(a+h),$$

or, substituting for λ_4 its value,

$$(a-h)[(a-h) - 8f\{\sqrt{(a)} - f\}] + 8(a+h)\{\sqrt{(a)} - f\}^2 = 0,$$

the condition is

$$\{3a + h - 4f\sqrt{(a)}\}^2 = 0, \quad \text{or} \quad 4f\sqrt{(a)} = 3a + h,$$

(observe that, f having been assumed negative, this implies $-h > 3a$). That is, $3a+h$ being $= -$ but not otherwise, the double tangent 4 will, for the value

$$-f = \frac{-3a-h}{4\sqrt{(a)}},$$

come to coincide with the line 2; and for any greater value of $-f$ will be as before above line 2, (being in this case an ordinary double tangent with real points of contact) as appears from the form, $U^2 = 0$, of the foregoing equation for the determination of f.

The passage of the line 3 to infinity, and the coincidence of the lines 4 and 2 may take place for the same value of f, viz. this will be the case if

$$\sqrt{(a)} + (a-h) = \frac{-3a-h}{4\sqrt{(a)}},$$

that is, if

$$7a + 4\sqrt{(a)} + h\{1 - 4\sqrt{(a)}\} = 0 \quad \text{or} \quad -h = \frac{a\{7 + 4\sqrt{(a)}\}}{1 - 4\sqrt{(a)}};$$

or, a being small, for the value $-h = 7a$ approximately. If $-h$ is less than the above value, then

$$\frac{-3a-h}{4\sqrt{(a)}}$$

is less than $\sqrt{(a)} + a - h$, or $-f$ increasing from $\sqrt{(a)}$, the coincidence of the lines 2, 4 takes place before the line 3 goes off to infinity: contrarywise, if $-h$ is greater than the above value.

In any form of the curve (i.e. whatever be the value of f in regard to a, h), if we imagine a, h indefinitely diminished, the lines 1, 2 and 4 will continually approach C, and the curve will gather itself up into certain definite portions of the lines $x = 0$, $y = 0$. Thus any secant through A (not being indefinitely near to the line AC), which meets the curve in real points, will meet it in two points tending to coincide at the intersection of the secant with the line $x = 0$; analytically there are always two intersections real or imaginary which (the secant not being indefinitely near the line AC) tend to coincide at the intersection of the secant with the line $x = 0$; and we thus see how we ultimately arrive at the line $x = 0$ twice repeated; and similarly for the line $y = 0$.